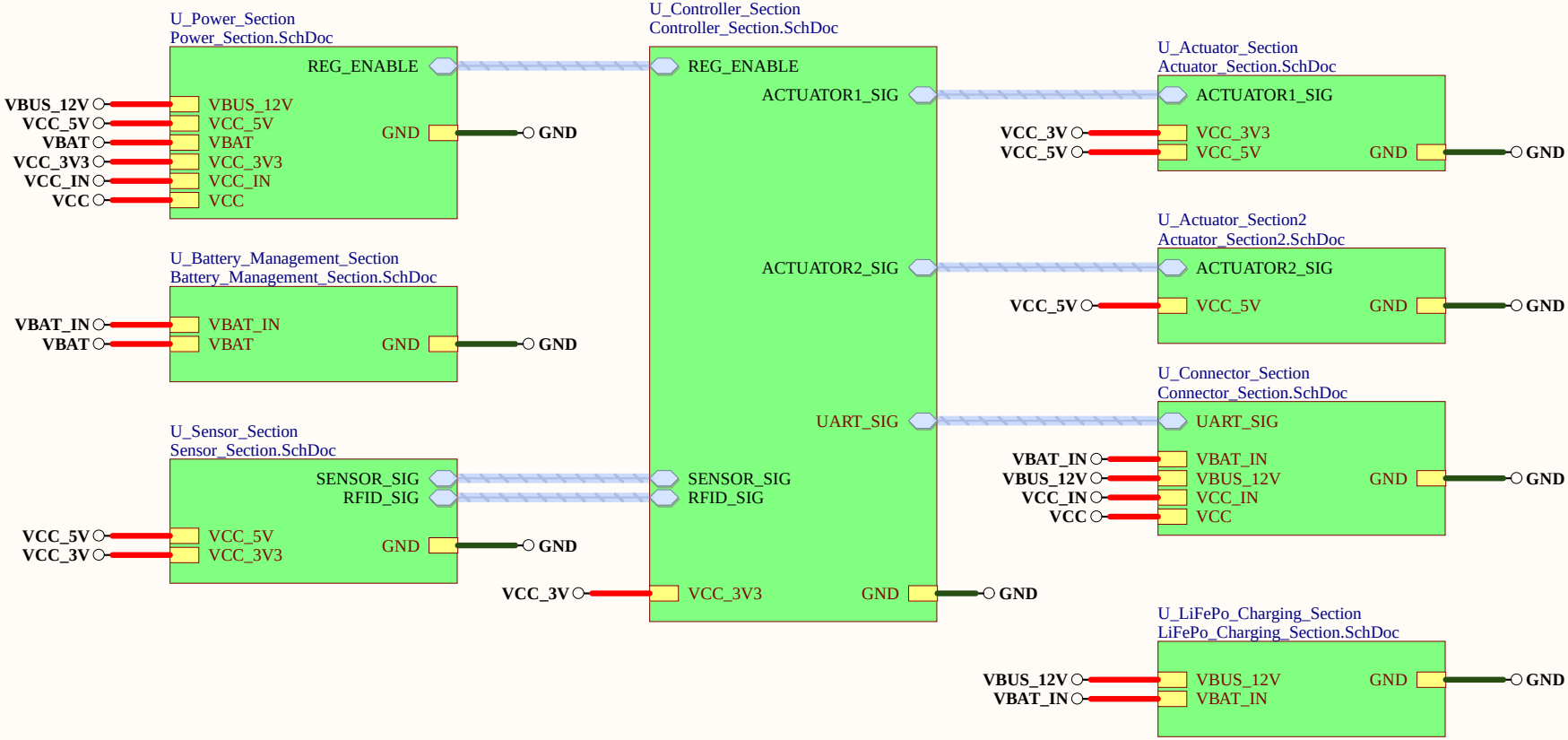
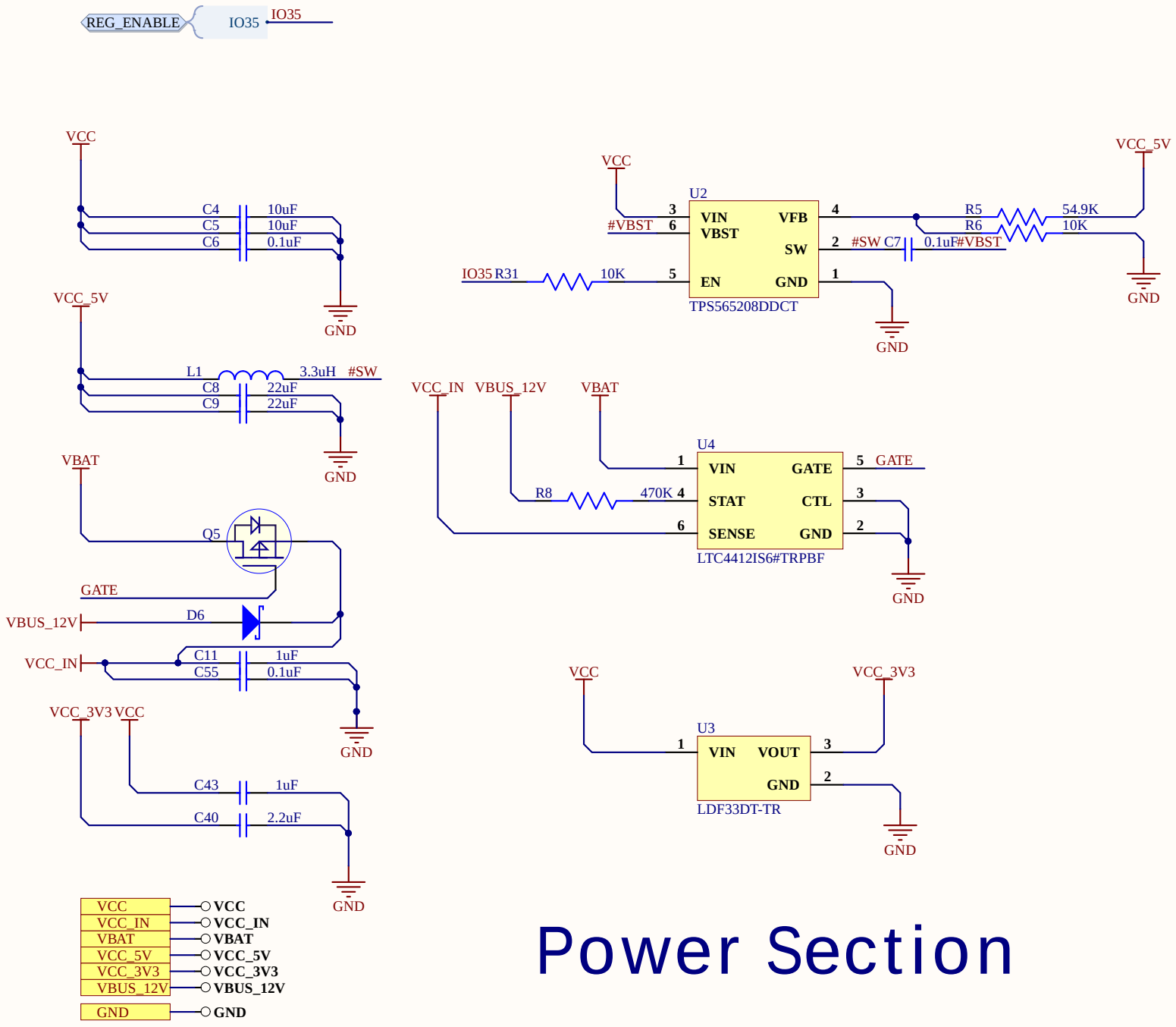
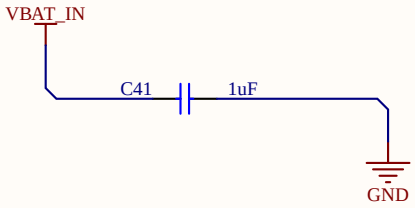
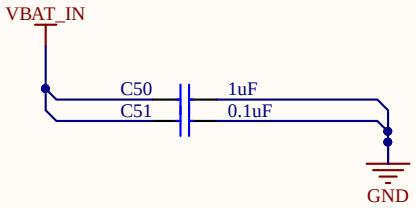
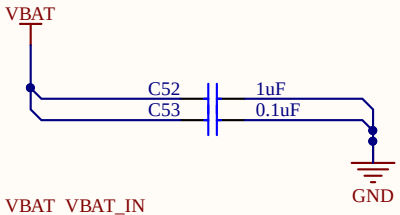
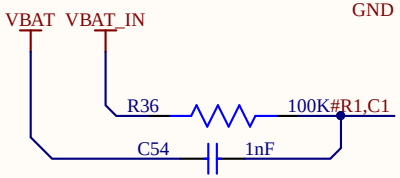

U_Power_Section
Power_Section.SchDoc
REG_ENABLE
VBUS_12V
VCC_5V
VBAT
VCC_3V3
VCC_IN
VCC
GND
U_Battery_Management_Section
Battery_Management_Section.SchDoc
VBAT_IN
VBAT
GND
U_Sensor_Section
Sensor_Section.SchDoc
SENSOR_SIG
RFID_SIG
VCC_5V
VCC_3V3
GND
U_Controller_Section
Controller_Section.SchDoc
REG_ENABLE
ACTUATOR1_SIG
ACTUATOR2_SIG
UART_SIG
SENSOR_SIG
RFID_SIG
VCC_3V3
GND
U_Actuator_Section
Actuator_Section.SchDoc
ACTUATOR1_SIG
VCC_3V3
VCC_5V
GND
U_Actuator_Section2
Actuator_Section2.SchDoc
ACTUATOR2_SIG
VCC_5V
GND
U_Connector_Section
Connector_Section.SchDoc
UART_SIG
VBAT_IN
VBUS_12V
VCC_IN
VCC
GND
U_LiFePo_Charging_Section
LiFePo_Charging_Section.SchDoc
VBUS_12V
VBAT_IN
GND

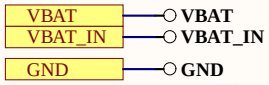

REG_ENABLE
IO35
IO35
VCC
C4 10uF
C5 10uF
C6 0.1uF
GND
VCC_5V
L1 3.3uH #SW
C8 22uF
C9 22uF
GND
VBAT
Q5
GATE
VBUS_12V
D6
VCC_IN
C11 1uF
C55 0.1uF
GND
VCC_3V3 VCC
C43 1uF
C40 2.2uF
GND
VCC
VCC_IN
VBAT
VCC_5V
VCC_3V3
VBUS_12V
GND
VCC
VCC_IN
VBAT
VCC_5V
VCC_3V3
VBUS_12V
GND
VCC
U2
3 VIN
6 VBST
4 VFB
2 SW
5 EN
1 GND
#VBST
IO35 R31 10K
TPS565208DDCT
#SW C7 0.1uF
#VBST
R5 54.9K
R6 10K
VCC_5V
GND
GND
VCC_IN VBUS_12V VBAT
U4
1 VIN
4 STAT
6 SENSE
5 GATE
3 CTL
2 GND
R8 470K
LTC4412IS6#TRPBF
GND
VCC
VCC_3V3
U3
1 VIN
3 VOUT
2 GND
LDF33DT-TR
GND
Power Section

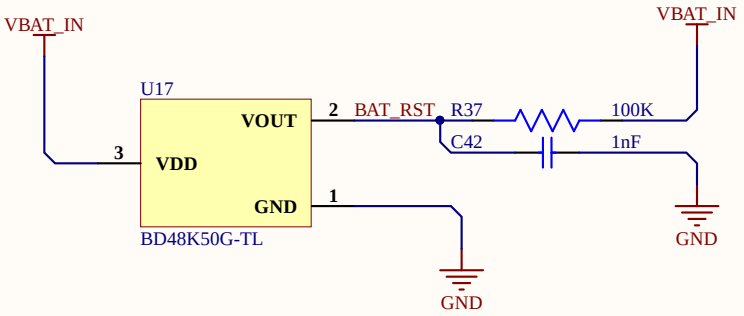


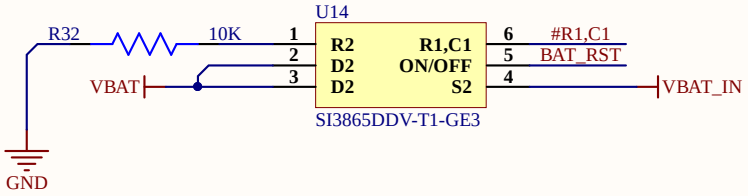




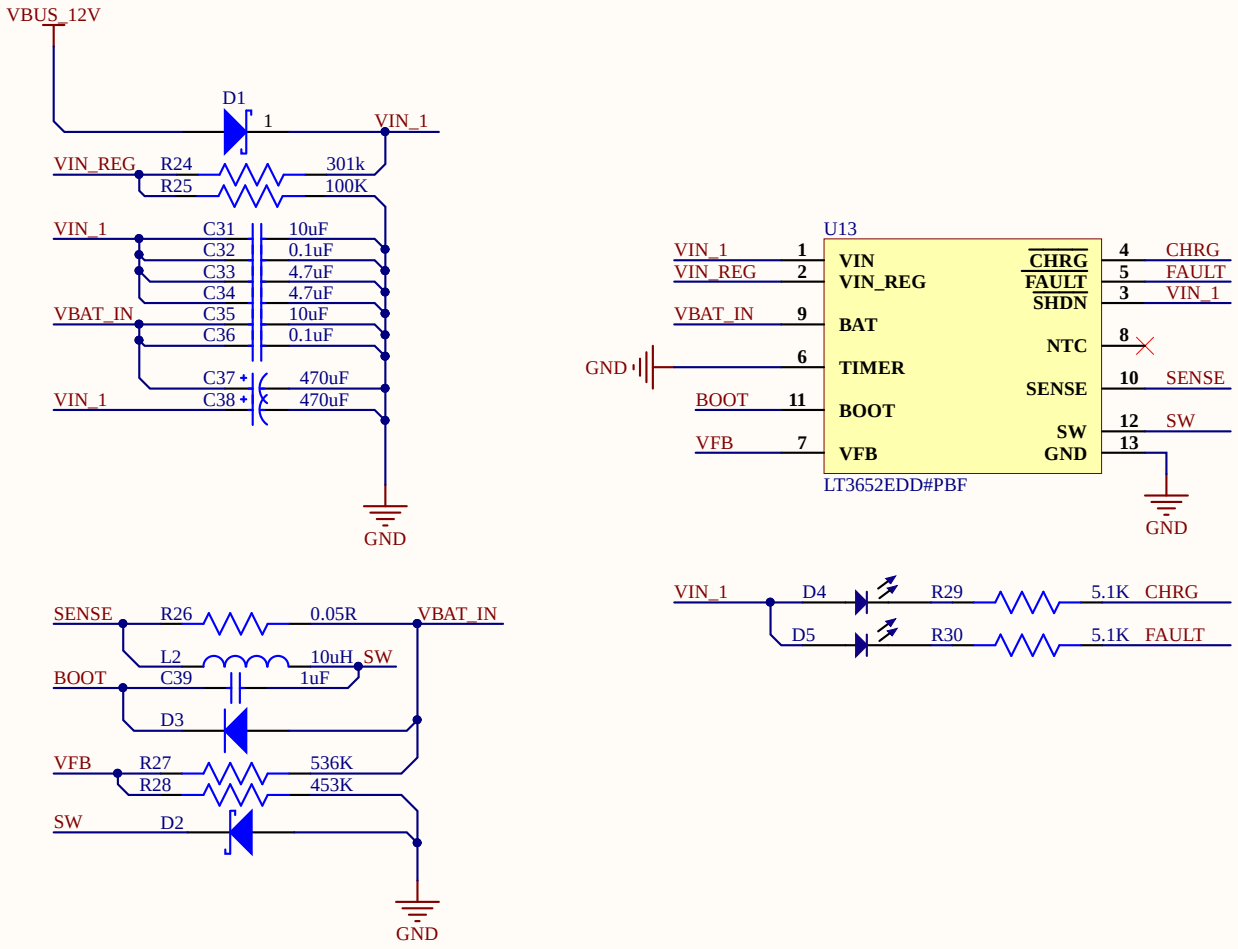






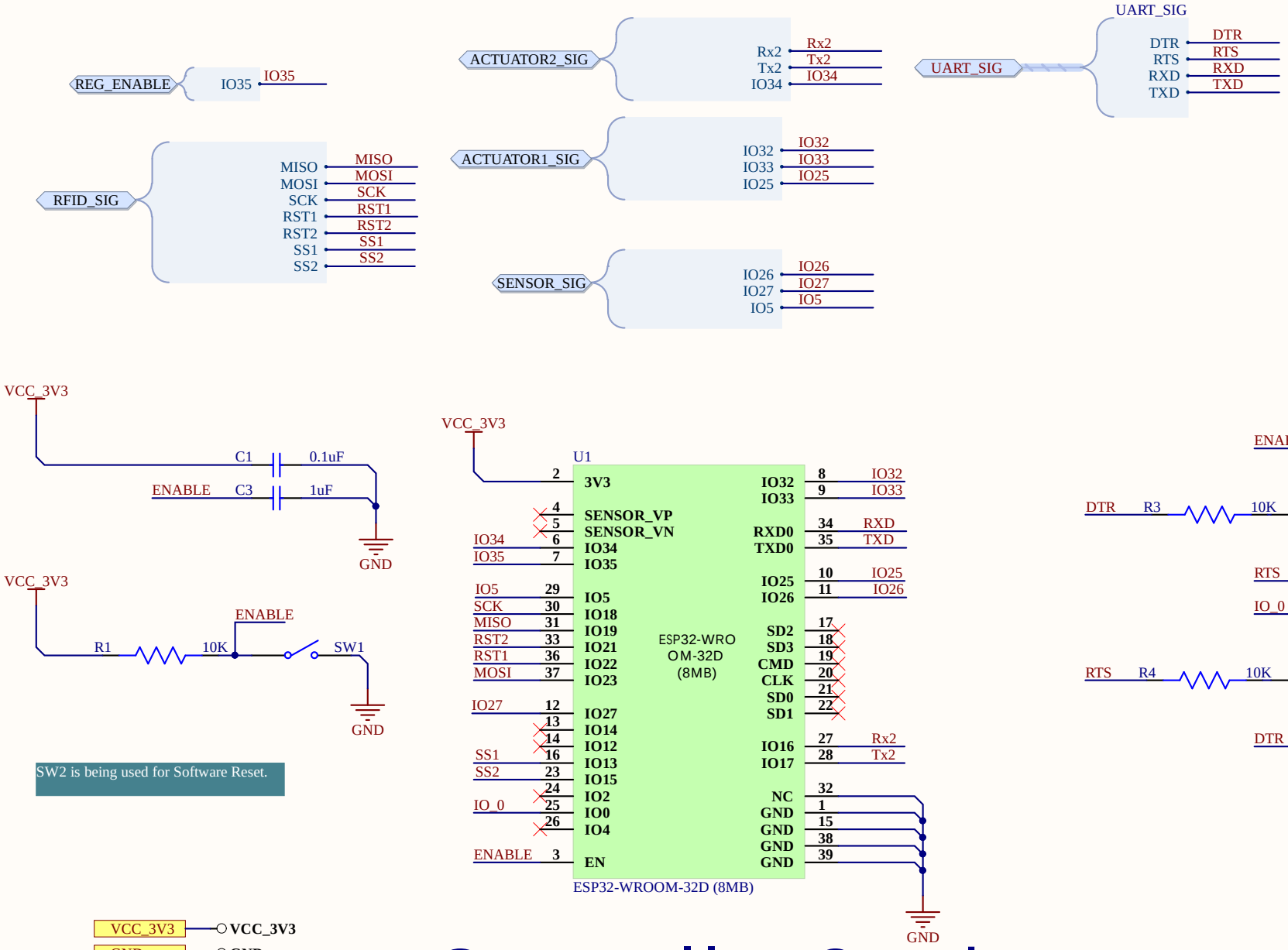


Battery Management Section

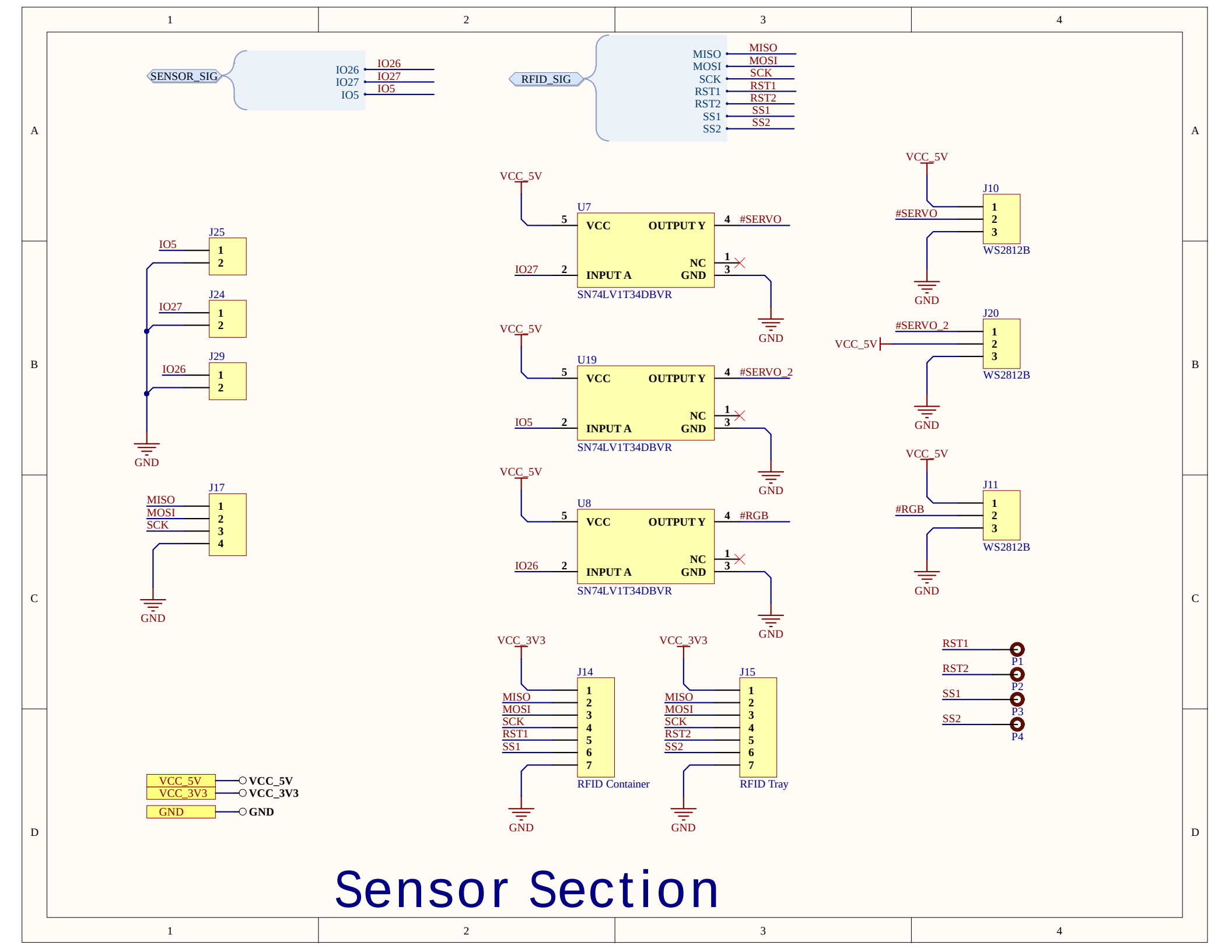


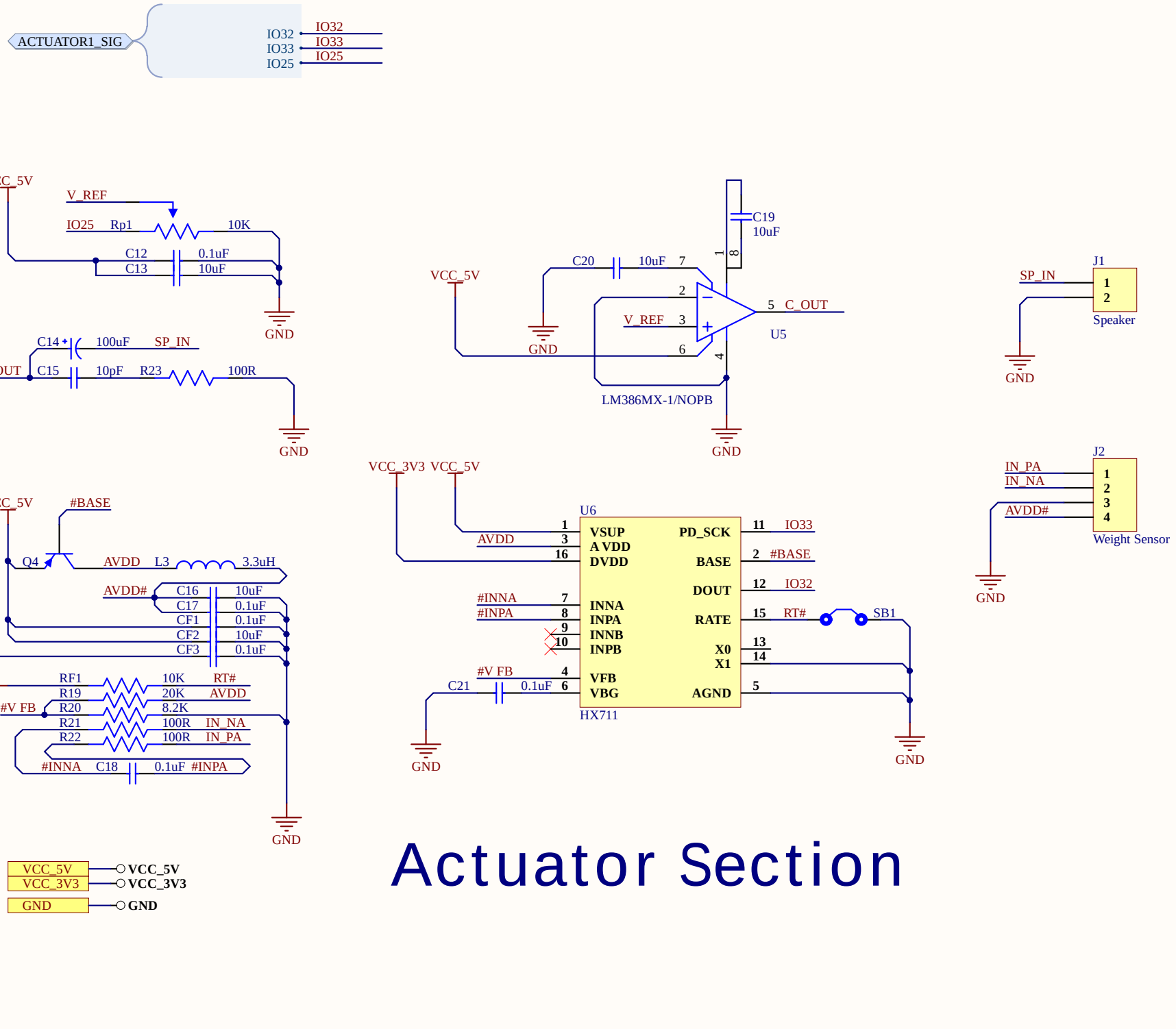
VBUS_12V	VBUS_12V
VBAT_IN	VBAT_IN
GND	GND

LiFePO4 Charging Section

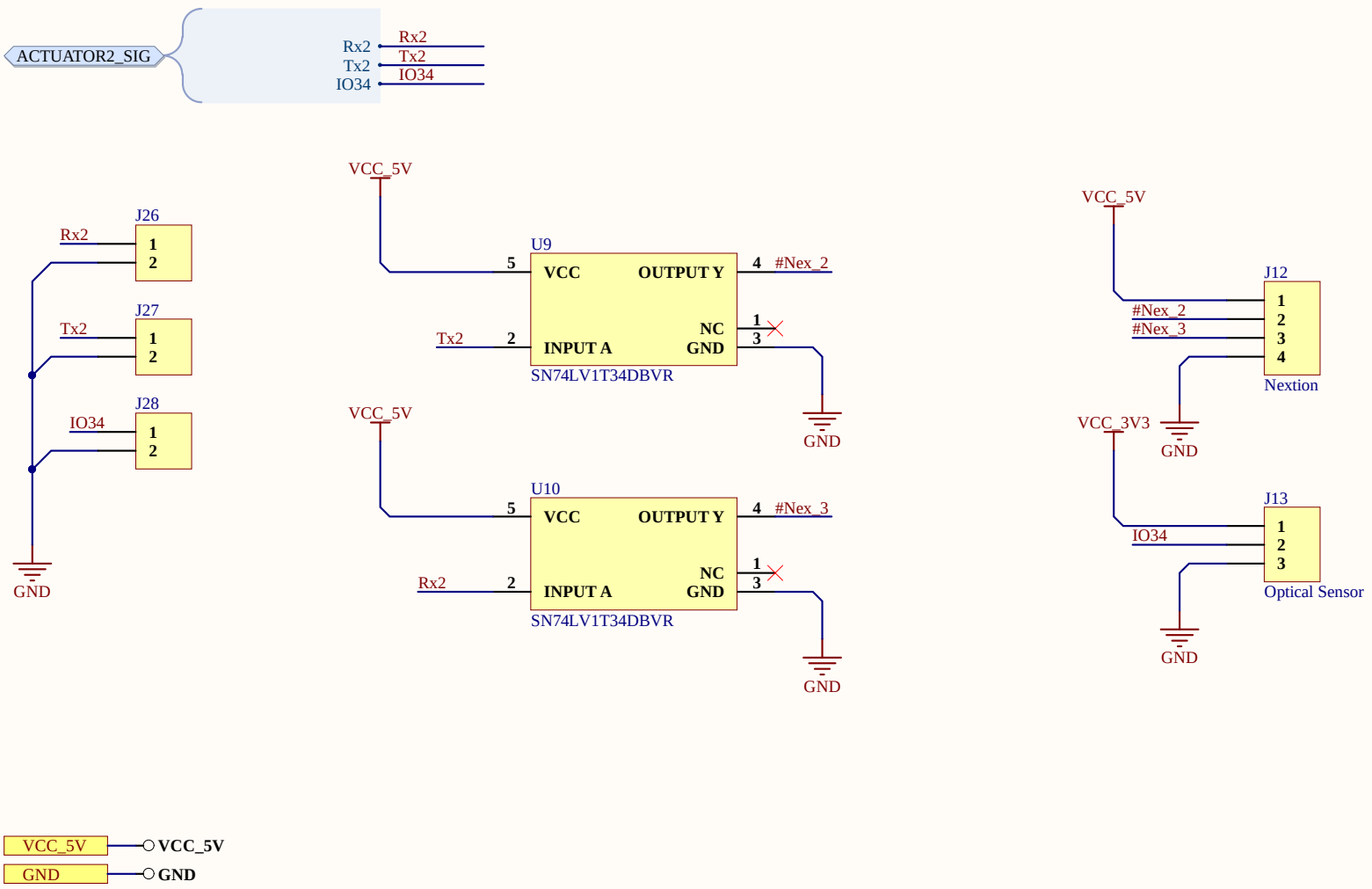


Controller Section

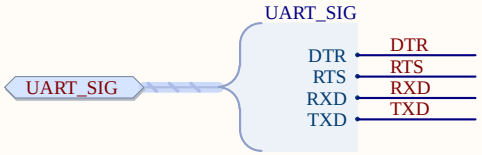

SENSOR_SIG
IO26
IO27
IO5
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
J25
J24
J29
J17
GND
U7
U19
U8
SN74LV1T34DBVR
VCC
OUTPUT Y
INPUT A
NC
#SERVO
#SERVO_2
#RGB
VCC_5V
J10
J20
J11
WS2812B
VCC_3V3
J14
J15
RFID Container
RFID Tray
P1
P2
P3
P4
VCC_5V
VCC_3V3
GND
Sensor Section

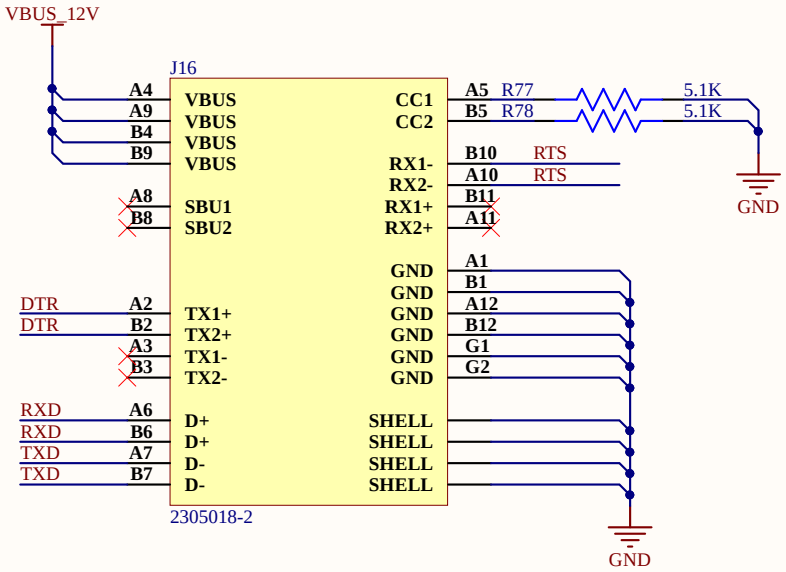


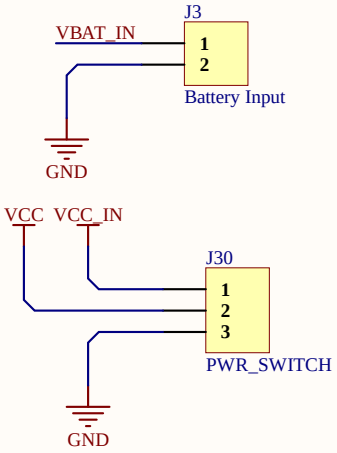
Actuator Section

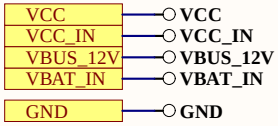


Actuator Section

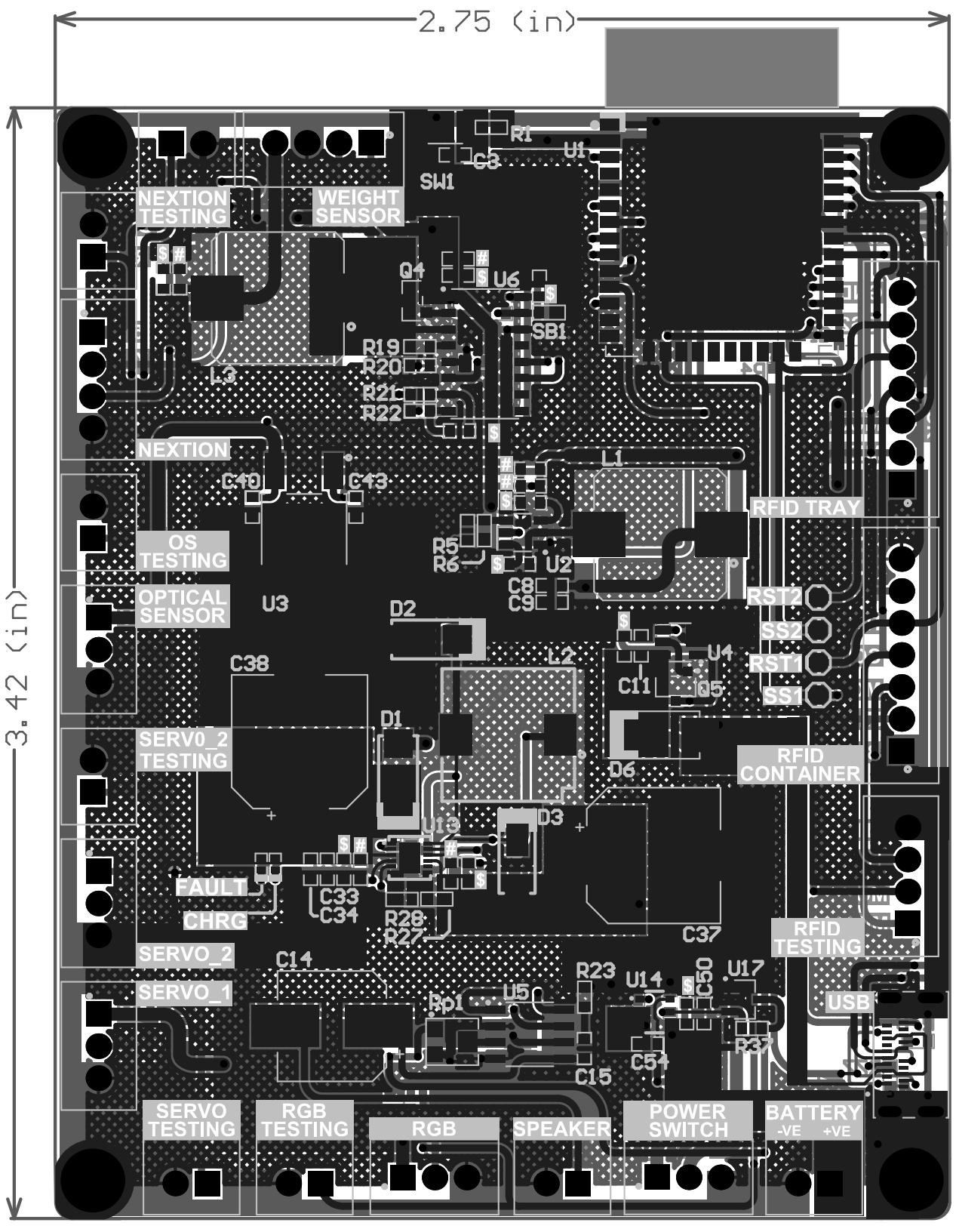








Connector Section


2.75 (in)
3.42 (in)
NEXTION TESTING
WEIGHT SENSOR
SW1
R1
C3
U1
Q4
U6
SB1
R19
R20
R21
R22
L3
NEXTION
C40
C43
L1
RFID TRAY
OS TESTING
R5
R6
U2
C8
C9
OPTICAL SENSOR
U3
D2
RST2
SS2
RST1
SS1
U4
Q5
C11
L2
C38
D1
SERV0_2 TESTING
RFID CONTAINER
D6
D3
U13
FAULT
CHRG
C33
C34
R28
R27
C37
RFID TESTING
SERVO_2
SERVO_1
C14
R23
U14
C50
U17
Rp1
U5
USB
R37
C54
C15
SERVO TESTING
RGB TESTING
RGB
SPEAKER
POWER SWITCH
BATTERY
-VE
+VE